

Nitrogen compounds

A Level Chemistry

Primary and secondary amines

An **amine** 胺 has an -NH_2 (primary) or -NH (secondary) group.

Making amines

- a **halogenoalkane** 卤代烷 with NH_3 in ethanol, heated under pressure $\rightarrow$ a primary amine.
- a halogenoalkane with a primary amine, heated under pressure $\rightarrow$ a secondary amine.
- **reduction** 还原 of an **amide** 酰胺 with LiAlH_4 .
- reduction of a **nitrile** 腈 with LiAlH_4 or H_2/Ni .

An amine (or **ammonia** 氨) reacts with an acyl chloride in a **condensation** 缩合 reaction to give an amide. The reagent is an **acyl chloride** 酰氯.

Basicity of amines

The **basicity** 碱性 of an amine comes from the lone pair on its nitrogen, which can accept an H^+ from water.

Phenylamine and azo compounds

Making phenylamine

Make **phenylamine** 苯胺 from benzene in two steps: nitrate benzene to nitrobenzene, then reduce it with hot Sn and concentrated HCl , followed by NaOH(aq) .

Reactions

- with bromine water at room temperature $\rightarrow$ 2,4,6-tribromophenylamine (a white precipitate). The ring is activated, like phenol.
- with HNO_2 (from NaNO_2 and dilute acid) below 10°C $\rightarrow$ a diazonium salt; warming this with water gives phenol.

Relative basicity



Ethylamine is the strongest base: its alkyl group pushes electron density onto the nitrogen, making the lone pair more available. Phenylamine is the weakest, because its lone pair is **delocalised** 离域 into the benzene ring, so it is less available to accept an H^+ .

Azo compounds

A **diazonium salt** 重氮盐 (benzenediazonium chloride) couples with **phenol** 苯酚 in NaOH(aq) to form an **azo compound** 偶氮化合物. The azo group is -N=N- . Azo compounds are brightly coloured and are often used as **dye** 染料 s; many other azo dyes are made the same way.

Amides

An amide is made from ammonia or a primary amine with an acyl chloride at room temperature.

Its reactions:

- **hydrolysis** with aqueous acid or alkali, giving the carboxylic acid (or its salt) and the amine (or ammonium).
- **reduction** of the C=O group with LiAlH_4 to give an amine.

An amide is a **much weaker base** than an amine, because the nitrogen lone pair is delocalised onto the neighbouring C=O group, so it is not available to accept an H^+ .

Amino acids

An **amino acid** 氨基酸 has both a basic $-\text{NH}_2$ group and an acidic $-\text{COOH}$ group.

Zwitterions and the isoelectric point

The $-\text{COOH}$ can give its H^+ to the $-\text{NH}_2$ in the same molecule, forming a **zwitterion** 两性离子 ($\text{H}_3\text{N}^+-\text{CHR}-\text{COO}^-$) —an ion with both a positive and a negative end but no overall charge.

- in acid (low pH), the amino acid gains H^+ and becomes **positive**.
- in alkali (high pH), it loses H^+ and becomes **negative**.
- at one special pH, the **isoelectric point** 等电点, it is mostly the zwitterion with no net charge.

Peptide bonds

Two amino acids join in a condensation reaction: the $-\text{COOH}$ of one and the $-\text{NH}_2$ of the other react, losing water and forming a **peptide bond** 肽键 (an amide link). Two amino acids give a **dipeptide** 二肽, three give a tripeptide.

Electrophoresis

In **electrophoresis** 电泳, a mixture is placed in an electric field at a chosen pH:

- above its isoelectric point, an amino acid is negative and moves to the positive electrode.
- below its isoelectric point, it is positive and moves to the negative electrode.
- at its isoelectric point, it does not move. So different amino acids separate.